Aiden Kim

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KAIST Aerospace Engineering undergraduate focused on guidance/control, cooperative engagement, real-time missile range estimation, flight simulation, and AI-assisted engineering systems. Interested in applying ML systems and agentic workflows to aerospace analysis, verification, and design automation.

Education

KAIST — Daejeon, Korea

B.S. in Aerospace Engineering *Expected Feb. 2028* Currently on a one-semester leave; returning next semester. Expected graduation after Fall 2027. Declared Aerospace Engineering in 2023.

Busan Science High School — Busan, Korea

Graduated, 17th cohort.

Awards and Recognition

- **Hanwha Aerospace Special Award**, Republic of Korea Air Force Academy Future Aerospace Academic Conference (2025).
- **Airbus Fly Your Ideas, Phase 2** (2026): In Vitro Firmware Red-Teaming for aerospace supply-chain security; mentored by Prof. Yongdae Kim (KAIST SSL).

Research Experience

KAIST Flight Dynamics and Control Lab (FDCL) — 2024–2026

Individual Study / Paid Undergraduate Researcher — *Prof. Chang Hoon Lee*

- Worked on real-time air-to-air missile range-estimation methods for no-escape-zone / launch-envelope visualization.
- Evaluated 60 Hz real-time feasibility; conventional scenarios retained substantial timing margin in simulation, while embedded aircraft hardware validation remained future work.

Undergraduate Research: PINNs for Aerospace Applications — 2025–2026

Physics-informed neural networks for guidance and trajectory estimation

- Explored neural and physics-informed approaches for aerospace guidance and trajectory-estimation problems.

Selected Projects

Cooperative Engagement and Guidance Simulation — Portfolio

Built a 3D/3-DoF Python engagement sandbox comparing PN, APN, and PIP-based cooperative guidance with aero limits, autopilot lag, thrust/mass burn, randomized target maneuvers, and parallel validation metrics.

Real-Time Missile Range Estimation — Research

Studied no-escape-zone / launch-envelope range estimation for air-to-air engagement visualization, with emphasis on real-time feasibility and computational margin under representative scenarios.

6-DOF and HVT Missile Defense Simulation — Portfolio

Developed MATLAB simulation workflows for high-value-target interception with NED dynamics, RK4 integration, phase-based PIP/PN guidance, maneuvering target models, batch comparison, and acceleration-effort analysis.

PINNs for Aerospace Guidance and Trajectory Generation — Portfolio

Proposed PINN-based real-time trajectory generation for 1-vs-1 air-to-air engagement using 3-DOF dynamics, receding-horizon inference, and PMP/HJB-style optimality constraints.

Agent Orchestration and Evaluation — Technical report

Designed and evaluated a hierarchical agent-orchestration harness using agent-assisted implementation workflows; studied harness amplification on SWE-bench Pro with Qwen3.6-27B.

Agent Infrastructure Tooling — Portfolio

Maintains tools for agent workflows, including semantic memory, code navigation, document transcription, web research, context compression, and task/project CLIs.

Other Work and Activities

- **Pathtent, Co-founder & CTO** (2026–): Pre-launch B2B SaaS for patent specification drafting and prior-art discovery workflows.
- **KAIST student leadership**: President of ASCEND (2024–2025), Vice President of Silver Lining (2023), Freshman Student Council design/planning (2022).

Technical Skills

Languages	Python, MATLAB, TypeScript, Bash, LaTeX
ML / Systems	PyTorch, MLX, vLLM, Modal, FastAPI, PostgreSQL, React / Next.js
Domains	Guidance and control, cooperative engagement, missile range estimation, flight dynamics, agent orchestration, evaluator pipelines